

Integrate Git with Docker for Source Controlled application build

Step 1 - Initialize a node application

- Run `npm init` to initialize a node app
- Run `npm i express` to install express

```
1RV24MC038_hrshreeraksha@asus:~/lab4$ npm init -y
Wrote to /home/1RV24MC038_hrshreeraksha/lab4/package.json:

{
  "name": "lab4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
1RV24MC038_hrshreeraksha@asus:~/lab4$ npm i express

added 68 packages, and audited 69 packages in 5s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step 2 - Create and write Dockerfile

- Run `touch Dockerfile`
- Insert the following commands into the docker file

```

1RV24MC038_hrshreeraksha@asus:~/lab4$ nano Dockerfile
1RV24MC038_hrshreeraksha@asus:~/lab4$ cat Dockerfile
#Build stage
FROM node:latest as build

WORKDIR /app

COPY package*.json ./
RUN npm install

COPY . .

#Stage 2
FROM node:latest

COPY --from=build /app /app

EXPOSE 5000

CMD ["node", "server"]

```

Step 3 - Create and write server.js

- Run *touch server.js* and Insert the lines of code into the file

```

1RV24MC038_hrshreeraksha@asus:~/lab4$ cat server.js
const express = require("express")
const app = express()
const PORT = 5000

app.get("/", (req,res)=>{
    res.send("Output of LAB 4 - Integrating git with Docker")
})

app.listen(PORT, ()=>console.log("Listening on", PORT));

```

Step 4 - Build the Docker file

- Run the following command *sudo docker build -t lab3-node .*

```

1RV24MC038_hrshreeraksha@asus:~/Lab4$ sudo docker build -t lab4-node .
[+] Building 1.5s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 234B                               0.0s
=> [internal] load metadata for docker.io/library/node:latest     0.4s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 43.71kB                                0.0s
=> CACHED [build 1/5] FROM docker.io/library/node:latest@sha256:7478f3725ef76ce6ba257a6818ea43c5eb7eb5bd424f0c 0.0s
=> CACHED [build 2/5] WORKDIR /app                                0.0s
=> CACHED [build 3/5] COPY package*.json ./                      0.0s
=> CACHED [build 4/5] RUN npm install                             0.0s
=> [build 5/5] COPY . .                                          0.3s
=> [stage-1 2/2] COPY --from=build /app /app                     0.1s
=> exporting to image                                             0.1s
=> => exporting layers                                           0.1s
=> => writing image sha256:94bb9772f50cae1dcc420fc7067e4d7776c8ea2031d52d74e911dd1835f88dd 0.0s
=> => naming to docker.io/library/lab4-node                      0.0s
1RV24MC038_hrshreeraksha@asus:~/Lab4$ █

```

Step 5 - Login to github

- Link your device with github through ssh by running the following commands

```

1RV24MC038_hrshreeraksha@asus:~/Lab4$ ssh-keygen -t ed25519 -C "shreerakshahr@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/1RV24MC038_hrshreeraksha/.ssh/id_ed25519):
Created directory '/home/1RV24MC038_hrshreeraksha/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/1RV24MC038_hrshreeraksha/.ssh/id_ed25519
Your public key has been saved in /home/1RV24MC038_hrshreeraksha/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:KdZnR1Y1SxtPxfStaeXNZJ4Nv0ahFMjFr9UP+xuhT/s shreerakshahr@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      .+o*=|
|      o o..B|
|      o.oX|
|      . . o. o/O|
|      o S o .**X|
|      . . o . o+ +|
|      . * |
|      + +|
|      +E|
+----[SHA256]-----+
1RV24MC038_hrshreeraksha@asus:~/Lab4$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIGVKHHpoKRBz90Qq9clIvLT6XshHW0UxgFs8z+ga3w1g shreerakshahr@gmail.com
1RV24MC038_hrshreeraksha@asus:~/Lab4$ ssh -T git@github.com
The authenticity of host 'github.com (64:ff9b::14cf:4952)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdKr4UvC0qU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi shree-raksha-hr! You've successfully authenticated, but GitHub does not provide shell access.

```

- Add your github username and email

```

1RV24MC038_hrshreeraksha@asus:~/Lab4$ git config --global user.email "shreerakshahr@gmail.com"
1RV24MC038_hrshreeraksha@asus:~/Lab4$ git config --global user.name "shree-raksha-hr"

```

Step 6 - Initialize a new git repository and commit changes

- To initialize a new git repository run *git init*

```
1RV24MC038_hrshreeraksha@asus:~/lab4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1RV24MC038_hrshreeraksha/lab4/.git/
1RV24MC038_hrshreeraksha@asus:~/lab4$ git add .
```

- Add and commit the changes by running *git add .* ; *git commit -m "Initial commit"*

```
1RV24MC038_hrshreeraksha@asus:~/lab4$ git commit -m "Initial commit"
[master (root-commit) 1d71222] Initial commit
636 files changed, 72893 insertions(+)
create mode 100644 Dockerfile
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/HISTORY.md
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
create mode 100644 node_modules/body-parser/package.json
create mode 100644 node_modules/bytes/History.md
create mode 100644 node_modules/bytes/LICENSE
create mode 100644 node_modules/bytes/Readme.md
create mode 100644 node_modules/bytes/index.js
create mode 100644 node_modules/bytes/package.json
```

Step 7 - Create a new git repository on GitHub

github.com/new

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * shree-raksha-hr / Repository name * lab4

✓ Your new repository will be created as lab-4.
The repository name can only contain ASCII letters, digits, and the characters ., -, and _.

Great repository names are short and memorable. How about **cuddly-dollop**?

Description

0 / 350 characters

2 Configuration

Choose visibility * Public

Choose who can see and commit to this repository

Add README Off

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license

Licenses explain how others can use your code. [About licenses](#)

Create repository

Step 8 - Add this new repository as the origin and push your code

```
1RV24MC038_hrshreeraksha@asus:~/lab4$ git branch -M main
1RV24MC038_hrshreeraksha@asus:~/lab4$ git remote add origin git@github.com:shree-raksha-hr/Lab4.git
1RV24MC038_hrshreeraksha@asus:~/lab4$ git push -u origin main
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 8 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 725.26 KiB | 2.56 MiB/s, done.
Total 719 (delta 139), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (139/139), done.
To github.com:shree-raksha-hr/Lab4.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

Step 9 - The files can be viewed on the created repository

The screenshot shows a GitHub repository page for 'shree-raksha-hr / Lab4'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is 'main'. The repository contains an initial commit from 'shree-raksha-hr' 36 minutes ago, which includes files: 'node_modules', 'Dockerfile', 'package-lock.json', 'package.json', and 'server.js'. The 'About' section is empty. The 'Releases' section shows no releases published. The 'Packages' section shows no packages published. The 'Languages' section shows a bar chart with JavaScript at 52.5% and Dockerfile at 47.5%. The 'Suggested workflows' section shows a suggestion for 'Webpack' based on the tech stack, with a 'Configure' button. The 'README' section is empty, with a prompt to 'Add a README'.

github.com/shree-raksha-hr/Lab4

shree-raksha-hr / Lab4

Code Issues Pull requests Actions Projects Wiki Security Insights

Lab4 Public Pin Watch 0 Fork 0 Star 0

main Go to file + <> Code About

shree-raksha-hr Initial commit 1d71222 · 36 minutes ago

File	Commit	Time
node_modules	Initial commit	36 minutes ago
Dockerfile	Initial commit	36 minutes ago
package-lock.json	Initial commit	36 minutes ago
package.json	Initial commit	36 minutes ago
server.js	Initial commit	36 minutes ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

Add a README

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

JavaScript 52.5%

Dockerfile 47.5%

Suggested workflows

Based on your tech stack

Webpack Configure

Build a NodeJS project with npm and webpack.